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Midterm Examination - 3

Phys332: Electromagnetic Theory II

2026/05/13

Please carefully read below before proceeding!

I acknowledge by taking this examination that I am aware of all academic honesty conducts that govern this course and how they also apply for this examination. I therefore accept that I will not engage in any form of academic dishonesty including but not limited to cheating or plagiarism. I waive any right to a future claim as to have not been informed in these matters because I have read the syllabus along with the academic integrity information presented therein.

I also understand and agree with the following conditions:

(1) all calculations are to be conducted in the notations and conventions of the formulae sheets provided during the exam unless explicitly stated otherwise in the question;

(2) I take *full responsibility* for any ambiguity in my selections in “multiple choice questions”;

(3) incorrect selections will receive $-1/7$ of the question’s points;

(4) I am expected to provide *step-by-step explanation of how I solved the question* and am expected to do so *only within the answer boxes* provided with the questions: the explanation is supposed to be succinct, well-articulated, and correct both scientifically and mathematically;

(5) no partial credit is awarded for the explanations provided in the answer boxes;

(6) some questions of some students will be randomly selected for inspection: *a question (if selected for inspection) might be awarded negative points* if its explanation is incorrect or insufficient to get the correct answer, even if the correct option is selected;

(7) any page which does not contain *both my name and student id* may not be graded;

(8) any extra sheet that I may use are for my own calculations and will not be graded.

Signature: ______________________

Question:	1	2	3	4	5	Total
Points:	21	21	21	21	21	105

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All dimensionfull quantities are to be understood in SI units.

(For instance, $\boxed{\epsilon_0 \simeq 10^{-11}}$ and $\boxed{|\mathbf{B}| = 5}$ stand for $\boxed{\epsilon_0 \simeq 10^{-11}\ \mathrm{A}^2\ \mathrm{kg}^{-1}\ \mathrm{m}^{-3}\ \mathrm{s}^4}$ and $\boxed{|\mathbf{B}| = 5\ \mathrm{A}^{-1}\ \mathrm{kg}^1\ \mathrm{s}^{-2}}$.)

Question: 1: Anisotropy in Linear Medium . . . . *(21 points)*

(Midterm Two Question Reloaded)

Consider a linear homogeneous medium with permittivity and permeability tensors given as

$$\epsilon_{ij} \doteq \begin{pmatrix} 1+a & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \epsilon \quad , \quad \mu_{ij} \doteq \begin{pmatrix} 1 & 0 & 0 \\ 0 & (1+b)^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix} \mu \tag{1}$$

for some constants a, b, ϵ, and μ. Under the assumption that there is no free charges or free currents in the medium, we can derive the differential equation describing the x and y components of the electric field in certain cases as

$$\begin{aligned} \left((1+\alpha_1)\frac{\partial^2}{\partial x^2} + (1+\alpha_2)\frac{\partial^2}{\partial y^2} + (1+\alpha_3)\frac{\partial^2}{\partial z^2} + (1+\alpha_4)\epsilon\mu\frac{\partial^2}{\partial t^2}\right) E_x(\mathbf{r},t) =& 0 \qquad (\text{if } a = 1 \text{ and } b = 0) \\ \left((1+\alpha_1)\frac{\partial^2}{\partial x^2} + (1+\alpha_2)\frac{\partial^2}{\partial y^2} + (1+\alpha_3)\frac{\partial^2}{\partial z^2} + (1+\alpha_4)\epsilon\mu\frac{\partial^2}{\partial t^2}\right) E_y(\mathbf{r},t) =& 0 \qquad (\text{if } a = 0 \text{ and } b = 1) \end{aligned} \tag{2}$$

(a) **(10½ points)** Consider the case $(a, b) = (1, 0)$; what is the correct tuple $(\alpha_1, \alpha_2, \alpha_3, \alpha_4)$?

- ☐ $(0, 0, 0, -2)$
- ☐ $(0, 0, 0, -3)$
- ☐ $(1, 0, 0, 0)$
- ☐ $(1, 0, 0, -2)$
- ■ $(1, 0, 0, -3)$
- ☐ $(0, 1, 1, 0)$
- ☐ $(0, 1, 1, -2)$
- ☐ $(0, 1, 1, -3)$

(b) **(10½ points)** Consider the case $(a, b) = (0, 1)$; what is the correct tuple $(\alpha_1, \alpha_2, \alpha_3, \alpha_4)$?

- ■ $(0, 0, 0, -2)$
- ☐ $(0, 0, 0, -3)$
- ☐ $(1, 0, 0, 0)$
- ☐ $(1, 0, 0, -2)$
- ☐ $(1, 0, 0, -3)$
- ☐ $(0, 1, 1, 0)$
- ☐ $(0, 1, 1, -2)$
- ☐ $(0, 1, 1, -3)$

Solution 1.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

Let's start with the maxwell equations which is also given in the formulae sheet:

$$\nabla \cdot \mathbf{D} = \rho_f \ , \quad \nabla \cdot \mathbf{B} = 0 \ , \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \ , \quad \nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \tag{3}$$

We can write this in index notation using Einstein's summation convention (we also use the fact that there are no free charges or currents):

$$\partial_i D_i = 0 \ , \quad \partial_i B_i = 0 \ , \quad \epsilon_{ijk}\partial_j E_k = -\partial_t B_i \ , \quad \epsilon_{ijk}\partial_j H_k = \partial_t D_i \tag{4}$$

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We now use the fact that the medium is linear, i.e. $D_i = \epsilon_{i\ell}E_\ell$ and $H_k = (\mu^{-1})_{k\ell}B_\ell$:

$$\partial_i(\epsilon_{i\ell}E_\ell) = 0\ , \quad \partial_i B_i = 0\ , \quad \epsilon_{ijk}\partial_j E_k = -\partial_t B_i\ , \quad \epsilon_{ijk}\partial_j\left((\mu^{-1})_{k\ell}B_\ell\right) = \partial_t(\epsilon_{i\ell}E_\ell) \tag{5}$$

Since we are given that a and b are constants, ϵ and μ are simply constant tensors:

$$\epsilon_{i\ell}\partial_i E_\ell = 0\ , \quad \partial_i B_i = 0\ , \quad \epsilon_{ijk}\partial_j E_k = -\partial_t B_i\ , \quad \epsilon_{ijk}(\mu^{-1})_{k\ell}\partial_j B_\ell = \epsilon_{i\ell}\partial_t E_\ell \tag{6}$$

We will now proceed as usual (as we have also done in class), which is also illustrated in the book for the isotropic case: take the curl of Faraday's law, insert Ampere's and Gauss's laws inside to get the wave equation. But before doing that, for the complete sake of illustration purposes, let's see what these equations look like in matrix notation:

$$\begin{pmatrix}\frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\end{pmatrix}\begin{pmatrix}1+a & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}E_x\\ E_y\\ E_z\end{pmatrix} = 0\ , \quad \begin{pmatrix}\frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\end{pmatrix}\begin{pmatrix}B_x\\ B_y\\ B_z\end{pmatrix} = 0 \tag{7a}$$

$$\begin{pmatrix}0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y}\\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x}\\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0\end{pmatrix}\begin{pmatrix}E_x\\ E_y\\ E_z\end{pmatrix} = -\begin{pmatrix}\frac{\partial B_x}{\partial t}\\ \frac{\partial B_y}{\partial t}\\ \frac{\partial B_z}{\partial t}\end{pmatrix}\ , \quad \begin{pmatrix}0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y}\\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x}\\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & 1+b & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}B_x\\ B_y\\ B_z\end{pmatrix} = \begin{pmatrix}1+a & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\frac{\partial E_x}{\partial t}\\ \frac{\partial E_y}{\partial t}\\ \frac{\partial E_z}{\partial t}\end{pmatrix} \tag{7b}$$

Clearly, one can work with purely compact vector notation

$$\nabla \times \mathbf{E}\ , \tag{8}$$

with index notation

$$\epsilon_{ijk}\partial_j E_k\ , \tag{9}$$

or with matrix notation

$$\begin{pmatrix}0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y}\\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x}\\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0\end{pmatrix}\begin{pmatrix}E_x\\ E_y\\ E_z\end{pmatrix}\ . \tag{10}$$

Going back and forth between these notations is actually straightforward (and trivial once you get used to it); however, matrix notation is not that useful beyond second rank tensors and the compact vector notation is limited in all practical purposes to three dimensional space only (hence it is not that useful in special relativity). Therefore, we will proceed with index notation (as we have done in the class as well).

For part (a), (31) becomes

$$\partial_i E_i = -a\frac{\partial E_x}{\partial x}\ , \quad \partial_i B_i = 0\ , \quad \epsilon_{ijk}\partial_j E_k = -\partial_t B_i\ , \quad \epsilon_{ijk}\partial_j B_k = \epsilon\mu\left(\frac{\partial E_i}{\partial t} + a\delta_{ix}\frac{\partial E_x}{\partial t}\right) \tag{11}$$

where we used

$$\epsilon_{ij} = \left(\delta_{ij} + a\delta_{ix}\delta_{jx}\right)\epsilon\ , \quad (\mu^{-1})_{ij} = \mu^{-1}\delta_{ij} \tag{12}$$

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Let us now take the curl of Faraday's law and insert other equations as appropriate:

$$\begin{aligned}
0 =&\epsilon_{ijk}\partial_j E_k + \partial_t B_i \\
0 =&\epsilon_{mni}\partial_n \left(\epsilon_{ijk}\partial_j E_k + \partial_t B_i\right) \\
0 =&(\epsilon_{imn}\epsilon_{ijk})\partial_n\partial_j E_k + \partial_t(\epsilon_{mni}\partial_n B_i) \\
0 =&\partial_m(\partial_k E_k) - (\partial_j\partial_j)E_m + \partial_t\left[\epsilon\mu\left(\frac{\partial E_m}{\partial t} + a\delta_{mx}\frac{\partial E_x}{\partial t}\right)\right] \\
0 =&\partial_m\left[-a\frac{\partial E_x}{\partial x}\right] - \left[(\partial_j\partial_j) - \epsilon\mu\frac{\partial^2}{\partial t^2}\right]E_m + a\mu\epsilon\delta_{mx}\frac{\partial^2 E_x}{\partial t^2}
\end{aligned} \tag{13}$$

hence

$$\left((a+1)\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} - (a+1)\epsilon\mu\frac{\partial^2}{\partial t^2}\right)E_x(\mathbf{r},t) = 0 \tag{14}$$

For part (b), (31) becomes

$$\partial_i E_i = 0\ , \quad \partial_i B_i = 0\ , \quad \epsilon_{ijk}\partial_j E_k = -\partial_t B_i\ , \quad \epsilon_{ijk}\partial_j B_k - b\,\delta_{ix}\partial_z B_y + b\,\delta_{iz}\partial_x B_y = \epsilon\mu\partial_t E_i \tag{15}$$

where we used

$$\epsilon_{ij} = \delta_{ij}\epsilon\ , \quad (\mu^{-1})_{ij} = \mu^{-1}\left(\delta_{ij} + b\,\delta_{iy}\delta_{jy}\right) \tag{16}$$

Let us now take the curl of Faraday's law and insert other equations as appropriate:

$$\begin{aligned}
0 =&\epsilon_{ijk}\partial_j E_k + \partial_t B_i \\
0 =&\epsilon_{mni}\partial_n \left(\epsilon_{ijk}\partial_j E_k + \partial_t B_i\right) \\
0 =&(\epsilon_{imn}\epsilon_{ijk})\partial_n\partial_j E_k + \partial_t(\epsilon_{mni}\partial_n B_i) \\
0 =&\partial_m(\partial_k E_k) - (\partial_j\partial_j)E_m + \partial_t\left[\epsilon\mu\partial_t E_m + b\,\delta_{mx}\partial_z B_y - b\,\delta_{mz}\partial_x B_y\right] \\
0 =&0 - \left[(\partial_j\partial_j) - \epsilon\mu\frac{\partial^2}{\partial t^2}\right]E_m + b\,\delta_{mx}\frac{\partial^2 B_y}{\partial t\partial z} - b\,\delta_{mz}\frac{\partial^2 B_y}{\partial t\partial x}
\end{aligned} \tag{17}$$

hence

$$\left(\nabla^2 - \epsilon\mu\frac{\partial^2}{\partial t^2}\right)E_y(\mathbf{r},t) = 0 \tag{18}$$

Question: 2: Attenuation of EM Waves in Conductors . . . . *(21 points)*
(Midterm Two Question Reloaded)

Please take $\epsilon \simeq 5\epsilon_0 \simeq 5\times 10^{-11}$ and $\mu \simeq \mu_0 \simeq 10^{-6}$ for this question: this is a rather crude yet computationally helpful approximation.

Consider a metal block, of the shape right rectangular prism, of conductance σ, put on top of a level table. Choose the coordinate system such that the surface of the table is the $z = 0$ plane, and the facets of metal block are $z = 0$ (its facet touching the table), $z = 1$, $y = -1$, $y = 1$, $x = 0$, and $x = 1$.

Assume that we have waited sufficiently long so that the transient period has ended; in other words, $\mathbf{E}$−field satisfies the partial differential equation $\left(\nabla^2 - \mu\epsilon\frac{\partial^2}{\partial t^2} - \mu\sigma\frac{\partial}{\partial t}\right)\mathbf{E}(\mathbf{r},t) = 0$.

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(a) **(10½ points)** For the real positive constants d & ω and the functions $k_\pm := \sqrt{d^{-2} \pm \epsilon\mu\omega^2}$, which option below can describe the real $\mathbf{E}$–field inside the metal for $|x| \ll 1$ and for a positive σ? *Hint: d denotes the **skin depth** and we expect the dielectric behavior as d goes to infinity.*

☐ $e^{-x/d}\cos(k_+x+\omega t)$ ■ $e^{-x/d}\cos(k_+x-\omega t)$ ☐ $e^{-x/d}\cos(k_-x+\omega t)$ ☐ $e^{-x/d}\cos(k_-x-\omega t)$

☐ $e^{x/d}\cos(k_+x+\omega t)$ ☐ $e^{x/d}\cos(k_+x-\omega t)$ ☐ $e^{x/d}\cos(k_-x+\omega t)$ ☐ $e^{x/d}\cos(k_-x-\omega t)$

(b) **(10½ points)** If we measure that a monochromatic blue light ($\omega \simeq 4\times10^{15}$) inside the metal gets attenuated with a 100 nanometer skin depth ($d = 10^{-7}$), what would we calculate for the conductivity σ of this metal? *Hint: $\mathbf{E}$-field should satisfy the above partial differential equation.*

☐ 2×10^4 ☐ 5×10^4 ☐ 1×10^5 ■ 1.5×10^5 ☐ 2×10^5 ☐ 5×10^5 ☐ 1×10^6 ☐ 1.5×10^6

Solution 2.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

In part (a), we may first use the hint to deduce that all options with $\cos\left(\sqrt{d^{-2}-\epsilon\mu\omega^2}x \pm \omega t\right)$ *are incorrect as they can not remain real in the limit $d\to\infty$. We further note that we are near the $x=0$ facet of the metal as we are told that $|x| \ll 1$ (we are far away from the other facet, i.e. $x=1$); therefore, the field should get attenuated as we increase x (we are moving deeper in the metal), hence we need $e^{-x/d} = e^{-|x|/d}$.*

The two physical considerations above (correct dielectric limit and attenuation inside the metal) leaves us with two options:

$$\mathbf{E}_\pm(\mathbf{r},t) = \mathbf{E}_0 e^{-x/d}\cos\left(\sqrt{d^{-2}+\epsilon\mu\omega^2}x \pm \omega t\right) \tag{19}$$

We can further comment that we are describing a wave going into *the metal and getting attenuated, hence its wave vector* $\mathbf{k}$ *should have the direction* $\hat{x}$*. This actually uniquely fixes the sign as a wave moving in the direction* $\mathbf{k}$ *has the form* $\cos(-\mathbf{k}\cdot\mathbf{r}+\omega t)$*. This is sufficient to solve the problem; however, let us not use this heuristic argument and determine the sign directly using the given differential equation:*

$$0 = \left(\nabla^2 - \mu\epsilon\frac{\partial^2}{\partial t^2} - \mu\sigma\frac{\partial}{\partial t}\right)\mathbf{E}(\mathbf{r},t) \tag{20a}$$

$$= \left(\frac{\partial^2}{\partial x^2} - \mu\epsilon\frac{\partial^2}{\partial t^2} - \mu\sigma\frac{\partial}{\partial t}\right)\left[\mathbf{E}_0 e^{-x/d}\cos\left(\sqrt{d^{-2}+\epsilon\mu\omega^2}x \pm \omega t\right)\right] \tag{20b}$$

$$= \mu\omega\left(\sigma \pm \frac{2\sqrt{1+d^2\epsilon\mu\omega^2}}{d^2\mu\omega}\right)\mathbf{E}_0 e^{-x/d}\sin\left(\sqrt{d^{-2}+\epsilon\mu\omega^2}x \pm \omega t\right) \tag{20c}$$

Since all parameters are positive real numbers, the only way that this identity is satisfied is if

$$\mathbf{E}(\mathbf{r},t) = \mathbf{E}_0 e^{-x/d}\cos\left(\sqrt{d^{-2}+\epsilon\mu\omega^2}x - \omega t\right) \tag{21a}$$

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and if

$$\sigma = \frac{2\sqrt{1+d^2\epsilon\mu\omega^2}}{d^2\mu\omega} \tag{21b}$$

Equations (21) make perfect sense! As $d \to 0$ (hence conductor becoming an ideal conductor), $\mathbf{E}$ field inside goes to zero (thanks to $e^{-x/d}$) and the conductivity σ goes to infinity. Conversely, $\sigma \to 0$ requires $d \to \infty$ and this reduces $\mathbf{E}$ to $\mathbf{E}_0 \cos(kx - \omega t)$ for $k = \sqrt{\epsilon\mu}\omega$.

In part (b), we simply insert the numbers:

$$\begin{aligned}\sigma =& \frac{2\sqrt{1+d^2\epsilon\mu\omega^2}}{d^2\mu\omega}\\ =& \frac{2\sqrt{1+(10^{-7})^2(5\times10^{-11})(10^{-6})(4\times10^{15})^2}}{(10^{-7})^2(10^{-6})(4\times10^{15})}\\ =& \frac{2\sqrt{1+8}}{4\times10^{-5}} = 1.5\times10^5\end{aligned} \tag{22}$$

It is intriguing to further comment on our result: for a cylindrical metal wire of this conductivity, of radius 1 mm and of length 1 meter, we have the resistance

$$R = \frac{\ell}{A\sigma} = \frac{1}{\pi(10^{-3})^2\times1.5\times10^5} \simeq 2\Omega \tag{23}$$

Considering that many electronic circuits work with resistors in kΩ range, such a wire can indeed be approximated as an ideal wire.

Question: 3: Gauge Freedom in Electromagnetic Theory . . . . *(21 points)*

Please take $\epsilon_0 \simeq 10^{-11}$ and $\mu_0 \simeq 10^{-6}$ for this question.

Consider a scalar field V and a vector field $\mathbf{A}$ defined through the relations $\mathbf{B} = \nabla\times\mathbf{A}$, $\mathbf{E} = -\nabla V - \frac{\partial\mathbf{A}}{\partial t}$.

(a) **(7 points)** If we were to change $\mathbf{A}$ to $\mathbf{A} + (y+5t)\hat{x} + (x+2z)\hat{y} + (2y+3t)\hat{z}$ and V to $V+\alpha$, which option below for α would be sufficient for this change to be a gauge transformation?

■ $-5x-3z-2t$ ☐ $5x+3z$ ☐ $5y+3z$ ☐ $-5y-3z-2t$

☐ $-5x^2-3z^2-2t^2$ ☐ $5x^2+3z^2$ ☐ $5y^2+3z^2$ ☐ $-5y^2-3z^2-2t^2$

(b) **(7 points)** For $\mathbf{A} = (x+t+y)^2\hat{y} - 2xtz\hat{z}$, what would be V at $(x,y,z,t) = (1,2,3,5)$ in the standard Lorenz gauge? *Hint: As we have explicitly discussed in class, Lorenz gauge is the one in which the Maxwell differential equations for* $\mathbf{A}$ *and* V *become decoupled wave equations. Furthermore,* $\mathbf{A} = 0$ *with* $V = 0$ *is consistent in Lorenz gauge.*

☐ -2×10^{17} ☐ 2×10^{17} ☐ -3×10^{17} ☐ 3×10^{17} ☐ -2×10^{18} ☐ 2×10^{18} ■ -3×10^{18} ☐ 3×10^{18}

(c) **(7 points)** Consider a *modified* version of the Lorenz gauge under which

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- the Maxwell differential equations for $\mathbf{A}$ and V become decoupled wave equations;
- for $\mathbf{A} = 0$, we have $\lim_{t\to 0} \frac{\partial V}{\partial t} = \pi \times 10^{17}$.

If $V = \hat{x} \cdot \mathbf{A} = \hat{y} \cdot \mathbf{A} = 0$, what is $\lim_{z\to z_0} \frac{\partial(\hat{z}\cdot\mathbf{A})}{\partial z}$ for $z_0 = \pi^{-1} \times 10^{-17}$ in this gauge?

☐ -1 ☐ 1 ☐ $-\pi$ ■ π ☐ -10^{-17} ☐ 10^{-17} ☐ -10^{17} ☐ 10^{17}

Solution 3.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

In part (a), we use the fact that a gauge transformation should leave the observables $\mathbf{E}$ *and* $\mathbf{B}$ *invariant; clearly, this requires*

$$\mathbf{A} \to \mathbf{A} + \nabla\lambda \,, \quad V \to V - \frac{\partial \lambda}{\partial t} \tag{24}$$

for an arbitrary scalar field λ. *If we are given*

$$\nabla\lambda = (y + 5t)\hat{x} + (x + 2z)\hat{y} + (2y + 3t)\hat{z} \tag{25}$$

we see that

$$\lambda(x, y, z, t) = xy + 2yz + 3zt + 5tx + g(t) \tag{26}$$

for an arbitrary function g. *We then see that* α *should be simply*

$$\alpha = -\frac{\partial V}{\partial t} = -5x - 3z - g'(t) \tag{27}$$

In part (b), Let us use the given hint: if we start with the Gauss and Ampere laws

$$\nabla \cdot \mathbf{E} = \epsilon_0^{-1}\rho \,, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \epsilon_0\mu_0 \frac{\partial \mathbf{E}}{\partial t} \tag{28}$$

and insert the potentials,

$$\nabla \cdot \left(-\nabla V - \frac{\partial \mathbf{A}}{\partial t}\right) = \epsilon_0^{-1}\rho \,, \quad \nabla \times (\nabla \times \mathbf{A}) = \mu_0\mathbf{J} + \epsilon_0\mu_0 \frac{\partial}{\partial t}\left(-\nabla V - \frac{\partial \mathbf{A}}{\partial t}\right) \tag{29}$$

we end up with

$$\begin{aligned} \Box V - \frac{\partial}{\partial t}\left(\nabla \cdot \mathbf{A} + \mu_0\epsilon_0 \frac{\partial V}{\partial t}\right) &= -\,\epsilon_0^{-1}\rho \\ \Box \mathbf{A} - \nabla\left(\nabla \cdot \mathbf{A} + \mu_0\epsilon_0 \frac{\partial V}{\partial t}\right) &= -\,\mu_0\mathbf{J} \end{aligned} \tag{30}$$

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This is a decoupled wave equation for V only if

$$\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} = constant \tag{31}$$

As we are further told that $\mathbf{A} = 0$ with $V = 0$ is consistent in Lorenz gauge, we finally arrive at the precise form of the Lorenz gauge:

$$\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} = 0 \tag{32}$$

We can now answer the question:

$$\mathbf{A} = (x + t + y)^2 \hat{y} - 2xtz\hat{z} \tag{33}$$

means

$$\frac{\partial V}{\partial t} = -\frac{2(x+t+y) - 2xt}{\mu_0 \epsilon_0} \quad \to \quad V = -\frac{2(x+y)t + t^2 - xt^2}{\mu_0 \epsilon_0} \tag{34}$$

hence

$$V(x = 1, y = 2, z = 3, t = 5) = -\frac{2 \times 3 \times 5}{10^{-6} \times 10^{-11}} = -3 \times 10^{18} \tag{35}$$

In part (c), our derivation for part (b) applies up to the equation (31). We now need $\lim_{t \to 0} \frac{\partial V}{\partial t} = \pi \times 10^{17}$ for $\mathbf{A} = 0$, which fixes that constant hence

$$\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} = \pi \tag{36}$$

is our modified Lorenz gauge. Technically, this is actually gauge equivalent to the Lorenz gauge itself, but we do not need to know that for this question. If $V = \hat{x} \cdot \mathbf{A} = \hat{y} \cdot \mathbf{A} = 0$, this condition becomes

$$\frac{\partial \hat{z} \cdot \mathbf{A}}{\partial z} = \pi \quad \to \quad \hat{z} \cdot \mathbf{A} = \pi z + g(x, y, t) \tag{37}$$

for arbitrary function g, hence $\lim_{z \to z_0} \frac{\partial (\hat{z} \cdot \mathbf{A})}{\partial z} = \pi$ for any z_0.

Question: 4: Electromagnetic field of time dependent current . . . . *(21 points)*
(Modified version of Example 10.2 of 978-1-009-39775-9)

Please feel free to use the identity $\frac{\mathrm{d}}{\mathrm{d}x} \log\left(-b - 2x + 2\sqrt{a^2 + bx + x^2}\right) = -(a^2 + bx + x^2)^{-1/2}$.

Consider an electrically-neutral infinite straight wire in an otherwise empty space. Assume that there is no current on the wire until $t = 0$; and, at time $t = 0$, a current of constant magnitude 4π gets abruptly turned on. Mathematically, $I(t) = 4\pi\theta(t)$ for the Heaviside-theta function θ. Set up a coordinate system such that any point on the wire has $x = y = 0$ and the current flows in $\hat{z}$ direction.

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(a) **(10½ points)** In Lorenz gauge, we can derive the vector potential as

$$\mathbf{A}(\mathbf{r},t)=\mu_0\hat{z}\int_\alpha^\beta \frac{dz}{\sqrt{z^2+r^2+2Cz\mathbf{r}\cdot\hat{z}}} \quad \text{for} \begin{cases}\alpha = B\mathbf{r}\cdot\hat{z}-A\sqrt{(ct)^2-r^2+(\mathbf{r}\cdot\hat{z})^2}\\ \beta = \mathbf{r}\cdot\hat{z}+A\sqrt{(ct)^2-r^2+(\mathbf{r}\cdot\hat{z})^2}\end{cases} \tag{38}$$

What is the tuple (A,B,C)?

☐ $(-1,-1,-1)$ ☐ $(-1,-1,1)$ ☐ $(-1,1,-1)$ ☐ $(-1,1,1)$

☐ $(1,-1,-1)$ ☐ $(1,-1,1)$ ■ $(1,1,-1)$ ☐ $(1,1,1)$

(b) **(10½ points)** For an arbitrary unit vector $\hat{n}$, we can derive $\mathbf{A}(ct\hat{n},t)=\mu_0\hat{z}\log\left(\frac{1+A|\hat{n}\cdot\hat{z}|^C}{1+B|\hat{n}\cdot\hat{z}|^C}\right)$ in the Lorenz gauge. What is the tuple (A,B,C)?

☐ $(-1,-1,1)$ ☐ $(-1,-1,2)$ ☐ $(-1,1,1)$ ☐ $(-1,1,2)$

■ $(1,-1,1)$ ☐ $(1,-1,2)$ ☐ $(1,1,1)$ ☐ $(1,1,2)$

Solution 4.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

For part (a), we start by deriving the vector potential. In Lorenz gauge,

$$\mathbf{A}(\mathbf{r},t)=\frac{\mu_0}{4\pi}\hat{z}\int_{-\infty}^{\infty}\frac{I\left(t-\frac{|\mathbf{r}-z\hat{z}|}{c}\right)}{|\mathbf{r}-z\hat{z}|}dz \tag{39}$$

where we oriented our cordinate system such that the wire is in the $z-$direction. We can now insert the formula for I:

$$\mathbf{A}(\mathbf{r},t)=\mu_0\hat{z}\int_{-\infty}^{\infty}\frac{\theta\left(t-\frac{|\mathbf{r}-z\hat{z}|}{c}\right)}{|\mathbf{r}-z\hat{z}|}dz \tag{40}$$

If $ct<|\mathbf{r}-z\hat{z}|$, the argument of Heaviside-theta becomes negative, hence the integrand becomes zero, hence we are restricted to the integration range for $ct>|\mathbf{r}-z\hat{z}|$ which means

$$(ct)^2>r^2+z^2-2z\mathbf{r}\cdot\hat{z} \quad \Rightarrow \quad (ct)^2-r^2+(\mathbf{r}\cdot\hat{z})^2>(z-\mathbf{r}\cdot\hat{z})^2 \tag{41}$$

hence

$$\mathbf{r}\cdot\hat{z}+\sqrt{(ct)^2-r^2+(\mathbf{r}\cdot\hat{z})^2}>z>\mathbf{r}\cdot\hat{z}-\sqrt{(ct)^2-r^2+(\mathbf{r}\cdot\hat{z})^2} \tag{42}$$

meaning

$$\mathbf{A}(\mathbf{r},t)=\mu_0\hat{z}\int_{\mathbf{r}\cdot\hat{z}-\sqrt{(ct)^2-r^2+(\mathbf{r}\cdot\hat{z})^2}}^{\mathbf{r}\cdot\hat{z}+\sqrt{(ct)^2-r^2+(\mathbf{r}\cdot\hat{z})^2}}\frac{dz}{\sqrt{z^2+r^2-2z\mathbf{r}\cdot\hat{z}}} \tag{43}$$

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For part (b), we have

$$\mathbf{A}(ct\hat{n},t)=\begin{cases}\mu_0\hat{z}\displaystyle\int_0^{2ct\hat{n}\cdot\hat{z}}\frac{dz}{\sqrt{z^2+(ct)^2-z(2ct\hat{n}\cdot\hat{z})}} & \text{for } \hat{n}\cdot\hat{z}>0\\ \mu_0\hat{z}\displaystyle\int_{2ct\hat{n}\cdot\hat{z}}^{0}\frac{dz}{\sqrt{z^2+(ct)^2-z(2ct\hat{n}\cdot\hat{z})}} & \text{for } \hat{n}\cdot\hat{z}<0\end{cases}\tag{44}$$

both of which can be combined into a single equation (by changing dummy variable $z\to -z$ and integration orientation in the second case):

$$\mathbf{A}(ct\hat{n},t)=\mu_0\hat{z}\int_0^{2ct|\hat{n}\cdot\hat{z}|}\frac{dz}{\sqrt{z^2+(ct)^2-z(2ct|\hat{n}\cdot\hat{z}|)}}\tag{45}$$

We are given the relation

$$\frac{1}{\sqrt{a^2+bx+x^2}}=-\frac{\mathrm{d}}{\mathrm{d}x}\log\left(-b-2x+2\sqrt{a^2+bx+x^2}\right)\tag{46}$$

which we can integrate and use the fundamental theorem of calculus to get

$$\int_\alpha^\beta\frac{dx}{\sqrt{x^2+a^2+bx}}=\log\left(\frac{-b-2\alpha+2\sqrt{a^2+b\alpha+\alpha^2}}{-b-2\beta+2\sqrt{a^2+b\beta+\beta^2}}\right)\tag{47}$$

specifically

$$\int_0^{-b}\frac{dx}{\sqrt{x^2+a^2+bx}}=\log\left(\frac{-b+2a}{b+2a}\right)\tag{48}$$

Comparing this with (45), we conclude

$$\mathbf{A}(ct\hat{n},t)=\mu_0\hat{z}\log\left(\frac{2ct|\hat{n}\cdot\hat{z}|+2ct}{-2ct|\hat{n}\cdot\hat{z}|+2ct}\right)\tag{49}$$

which simplifies to

$$\mathbf{A}(ct\hat{n},t)=\mu_0\hat{z}\log\left(\frac{1+|\hat{n}\cdot\hat{z}|}{1-|\hat{n}\cdot\hat{z}|}\right)\tag{50}$$

Apart from the μ_0, which is only there for the particular choice of the unit system, the result is independent of c or t! Although it is harder to understand why with our 3-dimensional approach, this is natural from a 4-dimensional perspective: the result is purely geometric as it shows the so-called lightcone structure.

Question: 5: Analysis of radiation in the temporal gauge .... *(21 points)*
(in the conventions of Griffiths)

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Define radiation as the part of the electromagnetic fields that can carry energy to the spatial infinity. We then define *radiation power* at t as $P(t) := \lim_{R\to\infty} \oint_{S_R^2} \mathbf{S}(\mathbf{r}, t + |\mathbf{r}|/c) \cdot \mathbf{dA}$ for S_R^2 denoting a sphere of radius R where $\mathbf{S}(\mathbf{r}, t)$ is the Poynting vector at the position $\mathbf{r}$ and time t. In the so-called *temporal gauge*, we choose $V = 0$ under which an explicit derivation leads to the conclusion that the most dominant piece in $\mathbf{S}$ as r goes to infinity is

$$\mathbf{S}(\mathbf{r}, t_r) \supset \left[\left| \mathbf{A}'(\mathbf{r}, t_r) \right|^2 \hat{r} + \alpha (\mathbf{A}'(\mathbf{r}, t_r) \cdot \hat{r}) \mathbf{A}'(\mathbf{r}, t_r) \right] / c \tag{51}$$

for some constant α and for the shorthand notation $\mathbf{A}'(\mathbf{r}, t_r) := \dfrac{\partial \mathbf{A}(\mathbf{r}, t_r)}{\partial t_r}$, and that only this part can actually contribute to the radiation. Which statement below is then completely correct?

- ☐ We can now use $\mathbf{A}(\mathbf{r}, t) \propto \displaystyle\int \frac{\mathbf{J}(\mathbf{r}', \mathbf{t_r})}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}'$ to compute the total radiation power.
- ☐ An arbitrary time dependence of $\mathbf{A}$ would suffice for nonzero radiation.
- ■ **Only the transverse component of A can contribute to the radiation.**
- ☐ Radiation computed this way is the extreme of all possible radiations subject to the gauge choice.
- ☐ If we used Coulomb gauge instead of temporal gauge, we would find precisely the same functional form of $\mathbf{A}$.
- ☐ There is no residual gauge transformation left, as we completely fixed $V = 0$.
- ☐ Poynthing vector being quadratic in $\mathbf{A}$ is a special property of the temporal gauge, and is absent in other (for instance radiation) gauges.
- ☐ This whole question is incorrect, because our very definition of radiation power was done in Lorenz gauge, and then we tried to analyze it in a completely different gauge!

Solution 5.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

We are told that we should work in a gauge where $V = 0$: this is of course possible as we can see in (24)

$$\mathbf{A} \to \mathbf{A} + \nabla\lambda\,, \quad V \to V - \frac{\partial \lambda}{\partial t} \tag{52}$$

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that a suitable λ can always be found to make $V = 0$. In this gauge, we have

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}\,, \quad \mathbf{B} = \nabla \times \mathbf{A} \tag{53}$$

hence

$$\mathrm{S}(\mathbf{r}, t) = -\frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t} \times (\nabla \times \mathbf{A}(\mathbf{r}, t)) \tag{54}$$

For $t_r := t - \frac{|r|}{c}$ denoting the retarded time, we have

$$\mathrm{S}(\mathbf{r}, t_r) = -\frac{\partial \mathbf{A}(\mathbf{r}, t_r)}{\partial t_r} \times (\nabla \times \mathbf{A}(\mathbf{r}, t_r)) \tag{55}$$

Now, in general, the curl would act both on $\mathbf{r}$ and t_r; however, the first piece dies out faster at infinity hence can not contribute to the radiation. To understand the second piece, we note that

$$\frac{\partial \mathbf{A}(\mathbf{r}, t_r)}{\partial x^i} \supset \frac{\partial \mathbf{A}(\mathbf{r}, t_r)}{\partial t_r}\frac{\partial t_r}{\partial x^i} \quad \Rightarrow \quad \partial_i A_j(\mathbf{r}, t_r) \supset -\frac{1}{c}\frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r}\frac{x_i}{r} \tag{56}$$

hence

$$\begin{aligned} S_i(\mathbf{r}, t_r) &= -\epsilon_{ijk}\frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r}\epsilon_{k\ell m}\partial_\ell A_m(\mathbf{r}, t_r) \\ &\supset \frac{1}{c}\epsilon_{ijk}\frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r}\epsilon_{k\ell m}\frac{\partial A_m(\mathbf{r}, t_r)}{\partial t_r}\frac{x_\ell}{r} \\ &\supset \frac{1}{c}(\epsilon_{kij}\epsilon_{k\ell m})\frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r}\frac{\partial A_m(\mathbf{r}, t_r)}{\partial t_r}\frac{x_\ell}{r} \\ &\supset \frac{1}{c}\frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r}\frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r}\frac{x_i}{r} - \frac{1}{c}\frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r}\frac{\partial A_i(\mathbf{r}, t_r)}{\partial t_r}\frac{x_j}{r} \end{aligned} \tag{57}$$

yielding

$$\mathrm{S}(\mathbf{r}, t_r) \supset \frac{\hat{r}|\mathbf{A}'(\mathbf{r}, t_r)|^2 - (\mathbf{A}'(\mathbf{r}, t_r)\cdot\hat{r})\mathbf{A}'(\mathbf{r}, t_r)}{c} \tag{58}$$

for the shorthand notation $\mathbf{A}'(\mathbf{r}, t_r) := \frac{\partial \mathbf{A}(\mathbf{r}, t_r)}{\partial t_r}$. Let's now look at the given statements:

1. We can now use $\mathbf{A}(\mathbf{r}, t) \propto \int \frac{\mathbf{J}(\mathbf{r}', \mathrm{t_r})}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}'$ to compute the total radiation power.
   This formula is derived from wave equation for a decoupled $\mathcal{A}$ which was constructed thanks to the Lorenz gauge: it is not applicable in this different gauge!
2. An arbitrary time dependence of $\mathbf{A}$ would suffice for nonzero radiation.
   Not necessarily; in reality, the terms due to a single time derivative of the sources will cancel each other in their contribution to the radiation, we need the second order time derivatives (such as accelerated sources).

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3. Only the transverse component of $\mathbf{A}$ can contribute to the radiation.
   Look at the formula we have derived, (58): we see that part of the $\mathcal{A}$ that is parallel to $\hat{r}$ cancels between the two terms: longitudinal part of $\mathcal{A}$ does not contribute to the radiation!

4. Radiation computed this way is the extreme of all possible radiations subject to the gauge choice.
   Radiation is integral of Poynthing vector which only depends on $\mathbf{E}$, $\mathbf{B}$, and μ_0: all these three are gauge invariant quantities, hence this statement does not make sense!

5. If we used Coulomb gauge instead of temporal gauge, we would find precisely the same functional form of $\mathbf{A}$.
   Let's go over gauge conditions: temporal gauge ($V = 0$), Coulomb gauge ($\nabla \cdot \mathbf{A} = 0$), radiation gauge ($\nabla \cdot \mathbf{A} = \frac{\partial V}{\partial t} = 0$), and Lorenz gauge ($\nabla \cdot \mathbf{A} + \epsilon_0 \mu_0 \frac{\partial V}{\partial t} = 0$). Obviously, they are not independent; for instance, if we are in radiation gauge, then Lorenz gauge condition is already satisfied. However, temporal gauge itself does not imply Coulomb gauge, so there is no reason to make that claim!

6. There is no residual gauge transformation left, as we completely fixed $V = 0$.
   We know from (24) that $\mathbf{A} \to \mathbf{A} + \nabla\lambda$ with $V \to V - \dfrac{\partial \lambda}{\partial t}$ for any λ gives us a gauge transformation. If we choose λ to be time-invariant, then this gauge transformation preserves temporal gauge condition, hence it is a residual gauge transformation.

7. Poynthing vector being quadratic in $\mathbf{A}$ is a special property of temporal gauge, and is absent in other (for instance radiation) gauges.
   This is simply incorrect and can be quickly checked by dimensional analysis!

8. This whole question is incorrect, because our very definition of radiation power was done in Lorenz gauge, and then we tried to analyze it in a completely different gauge!
   Radiation power depends only on gauge-invariant quantities, hence its definition can not be taken specific for a particular gauge choice!

« « « Congratulations, you have made it to the end! » » »